

A strict computer-assisted refinement of a certified lower bound for simple critical-line zeros

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Abstract

We give a computer-assisted strict refinement of the supplied certified Gebendorfer lower bound for the lower asymptotic proportion of nontrivial zeros of the Riemann zeta function that are simple and lie on the critical line. The frozen endpoint is

$$\kappa_0^s \geq \frac{6734775921119}{10^{13}} = 0.6734775921119.$$

The exact difference from the supplied comparator is

$$\frac{34271814394247548721}{491155304728390000000000000} > 0.$$

An isolated hostile audit passed mathematical kill targets K1–K15. A final publication referee returned `PUBLICATION_PASS_WITH_EDITORIAL_DEFECTS_ONLY` with no substantive blocker. A later post-release regression audit found no mathematical or scope/trust-boundary blocker and triggered only public-layer self-containment, reproducibility, provenance, and rendering corrections in v0.1.1. The complete 58,577,037-node Gebendorfer finite layer was independently cross-host replayed. Partial Lean formalization is frozen with its remaining boundaries stated explicitly.

1 Statement and counting convention

Let $N(T)$ count all nontrivial zeros ρ of the Riemann zeta function with $0 < \text{Im } \rho \leq T$, counted with analytic multiplicity. Let $N_0^s(T)$ count the distinct zeros $\rho = \frac{1}{2} + i\gamma$ with $0 < \gamma \leq T$ whose analytic multiplicity is exactly one. Define

$$\kappa_0^s = \liminf_{T \rightarrow \infty} \frac{N_0^s(T)}{N(T)}.$$

The frozen theorem claim is

$$\boxed{\kappa_0^s \geq 0.6734775921119}.$$

The exact endpoint is $6734775921119/10^{13}$.

The supplied Gebendorfer comparator is

$$r_G = \frac{8269551442741204889710953}{12278882618209750000000000} = 0.673477522333941655\dots,$$

and the exact strict gain is

$$\frac{6734775921119}{10^{13}} - r_G = \frac{34271814394247548721}{49115530472839000000000000} > 0.$$

2 Imported certified local base

The imported local base is the supplied Gebendorfer six-gap architecture: one explicit analytic window, nine universal local inequalities, three-window compatibility, exact coarsening/joining, and a restricted eight-gap proof. The full finite computation comprises

$$50,089,182 + 7,638,774 + 849,081 = 58,577,037$$

visited nodes.

All computational obligations were independently replayed on a separate Windows host. The unmodified official `reproduce.py --full` returned code 1 at a raw-byte equality test after the nine local proofs because native Windows text mode emitted CRLF while the frozen artifact uses LF. A documented continuation completed the cover, coarsening, join, and restricted proof. After LF normalization the generated row sequences are identical, the 71-box coarsening is byte-identical, and the total node count matches the retained reference exactly.

3 Refinement architecture

The frozen post-Gebendorfer proof is organized into audited blocks K1–K15:

- K1 primary signed state, occurrence multiplicities, simple Gram and inertia;
- K2 one residual identity $D + Y$, with $Y = 2q_0 + W + J + Z$;
- K3 quotient/refinement with denominator boundary guards;
- K4 global rank/high-eigenvalue feasible region;
- K5 one nonlinear residual payment;
- K6 tail-aware profile and exact $\rho = 0$ regression;
- K7–K9 69 trim rows, physical capacity, $m = 492$ costs, $n = 486$, and 162 untrimmed triples;
- K10 464 exact chord/ray Farkas certificates for the universal supporting plane;
- K11 analytic transport, all offsets, endpoint losses, and fixed smoothing;
- K12 legal 13-module fusion with no double counting;
- K13 exact endgame with positive nonlinear margin;
- K14 independent exact reconstruction sealed before the author verifier was read;
- K15 disciplined public claim wording.

4 Full proof and evidence

This technical note is intentionally compact and is distributed with browseable proof/evidence plus GitHub Release assets. Key repository files are:

- `simple-zero-bound/proof/FUSION_PROOF.md`;
- `simple-zero-bound/proof/verify_fusion.py`;
- `simple-zero-bound/audit/INDEPENDENT_REDERIVATION.md`;
- `simple-zero-bound/audit/INDEPENDENT_FINAL_RECEIPT.json`;
- `simple-zero-bound/audit/FINAL_PUBLICATION_REFeree_REPORT.md`;
- `simple-zero-bound/reproducibility/PGF_C_REPLAY_QUICKSTART.md`;
- `simple-zero-bound/reproducibility/EXTERNAL_GEBENDORFER_INPUT.md`.

The v0.1.1 public-safe Release assets include the fusion/hostile derivatives, the PGF-C exact replay payload, cross-host replay evidence, and frozen Lean sources. Their SHA-256 pins are listed in `provenance/RELEASE_ASSET_HASHES.md`. The original third-party Gebendorfer reproduction ZIP and its exact extracted member files are not redistributed in the v0.1.1 public-safe asset set; the replay payload is hydrated locally from the external source after SHA-256 verification.

5 Audit status

The isolated hostile audit gave

`PASS_WITH_REPRODUCIBILITY_DEFECTS_ONLY`.

All K1–K15 mathematical kill targets passed; K16 was a technical default-path reproducibility issue.

The final publication referee gave

`PUBLICATION_PASS_WITH_EDITORIAL_DEFECTS_ONLY`

and found no substantive mathematical, scope, or circularity blocker. Its editorial findings and their RC3 resolutions are recorded in `simple-zero-bound/audit/REFeree_FINDINGS_RESOLUTION.md`.

6 Lean status

The release contains exact frozen Lean source/manifests for Stage 5, the Stage-6 bootstrap, and a substantive K1–K5 scalar/core tranche. The local research host reported build/axiom PASS statuses for the two Stage-6 tranches. The final referee did not rerun Lean and noted that raw build stdout was absent from RC2. Accordingly these are *reported local build statuses*, not independently reproduced build evidence.

This is partial formalization, not an end-to-end Lean proof. The concrete K1 Vandermonde–Sylvester zeta-operator realization remains an explicit boundary, K6–K15 are not fully internalized, and Lean does not rerun the 58,577,037-node interval computation.

7 Reproducibility

PGF-A supports relocating its frozen source archives with

```
python replay.py
--input-dir <directory-containing-input-zips>
```

The exact Windows continuation used after the official raw-byte failure is documented under `simple-zero-bound/reproducibility/`. For the exact finite fusion replay, extract `PGF_C_EXACT_REPLAY_PAYLOAD_V0_1_1.zip`, obtain the external Gebendorfer ZIP, and run

```
python -B -X utf8 hydrate_external_input.py
--gebendorfer-zip /path/to/Zeta_Reblocking_Repro.zip
python -B -X utf8 PGF-C/verify_fusion.py --all
```

The public package explicitly distinguishes the failed byte-identical single-process interval run from the successful cross-host continuation.

References

- [1] J. J. Gebendorfer, *Compatible local windows and a certified lower bound for simple critical-line zeros*, 7 September 2026. Manuscript/reproduction package is an external comparator input; see the repository external-input note for the source locator and SHA-256.
- [2] S. A. C. Baluyot, D. A. Goldston, A. I. Suriajaya, and C. L. Turnage-Butterbaugh, *An unconditional Montgomery theorem for pair correlation of zeros of the Riemann zeta-function*, *Acta Arithmetica* **214** (2024), 357–376. doi:10.4064/aa230612-20-3; arXiv:2306.04799v1.
- [3] Y. Lamzouri, *A new proof that more than 2/3 of the zeros of the Riemann zeta function are simple and on the critical line*, 2026. arXiv:2609.02882v1.

8 Scope

This is a computer-assisted strict refinement of the supplied certified comparator. It is not a proof of the Riemann Hypothesis, not full Lean formal verification, not completed independent human peer review, not a claim of global optimality, and not a literature-wide priority claim.